Shadow Removal of Hyperspectral Remote Sensing Images With Multiexposure Fusion

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***Abstract*— Shadow removal is a challenging problem in hyperspectral remote sensing images due to its spatial-variant properties and diverse patterns. In this work, a shadow removal framework with multiexposure fusion is proposed for hyperspectral remote sensing images, which consists of three major steps. First, a color space conversion method is exploited to detect the shadow regions. Second, the principle of the intrinsic decomposition model is utilized to generate a set of differently exposed hyperspectral images (HSIs), i.e., multiexposure images. Third, the generated multiexposure images and the original HSIs are fused together with a two-stage image fusion method so as to remove the shadows in hyperspectral remote sensing images effectively. Experiments performed on three real hyperspectral datasets confirm that the performance of the proposed method outperforms other state-of-the-art shadow removal approaches.**

***Index Terms*— Color space conversion, intrinsic image decomposition, multiexposure fusion, shadow removal, two-stage image fusion.**

I. Introduction

HYPERSPECTRAL images consist of narrow and continuous spectrums ranging from visible light to infrared wavelengths [1], [2], [3], [4]. Each object in hyperspectral image has a unique spectral curve, which helps for distinguishing different materials. Due to this advantage, hyperspectral images have been widely used in various aspects, such as land-cover classification [5], [6], [7], object detection [8], [9], [10], [11], and environment monitoring [12], [13], [14].

The illumination in hyperspectral imaging is often affected by shadow caused by cloud cover, buildings, and vegetation [15], [16], [17]. In this situation, the same material may have different spectral characteristics in the shadow area and nonshadow area [18]. This phenomenon has a serious impact on the following analysis of hyperspectral images [19], [20]. In order to address this problem, many shadow removal techniques have been developed, which mainly include two key steps: shadow detection and shadow removal [21], [22].

The objective of shadow detection is to segment the shadow regions accurately [23]. Currently, many studies have been developed to detect shadow regions [24]. For example, Rufenacht *et al.* [25] used a threshold method to detect shadow regions according to the low reflectivity characteristics of the shadow on the visible and near infrared (NIR) images. Han *et al.* [26] converted images from red-green-blue (RGB) to other color spaces (hue-intensity-saturation (HIS), hue-saturation-value (HSV), etc.) and constructed shadow indices using NIR components, hue values, and intensity values, and obtained shadow mask by automatically determining thresholds. The two methods are based on threshold segmentation method, and thus, are not able to obtain the optimal shadow detection effect in complex scenes [27]. Khan *et al.* [28] used multiple convolutional deep neural networks (ConvNets) to automatically learn the most relevant features of shadow to generate shadow contours. Nguyen *et al.* [29] developed a conditional generative adversarial network (cGAN) method to yield the shadow regions. However, most deep learning-based methods require real paired shadow/mask image datasets. The acquisition of a large number of accurate paired datasets is a challenging work [30].

Besides, many publications investigated shadow removal of RGB images [31], [32], [33], [34]. For example, Arbel and Hel-Or designed a shadow removal framework from a single image, in which the edges between shadow and nonshadow pixel intensities were removed to balance the illumination of the image [32]. Zhang *et al.* [33] constructed the correspondence between shadow and nonshadow patches based on texture similarity and built an optimized illumination recovering operator to restore the texture details of shadow patches. However, it is difficult to model the correlation between shadow regions and nonshadow regions in large complex scenes [35]. Liu *et al.* [34] proposed G2R-ShadowNet, which was jointly trained in an end-to-end scheme for shadow generation and shadow removal. Fu *et al.* [36] designed the shadow-aware FusionNet to produce fusion weight maps of all over-exposure

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images and utilized the boundary-aware RefineNet to remove shadow boundaries. These deep learning-based methods often require shadow/nonshadow image pairs to accurately learn model parameters [37], [38].

Compared with RGB images, shadow removal in hyperspectral images is also a challenging problem. The reason is that hyperspectral images have rich spectral information and complex spatial information. Recently, many methods have been proposed for removing shadows in hyperspectral images. For example, Liu *et al.* [39] applied the linear mixture model to recover the spectral reflectance of different objects in the shadow areas. Matsuki *et al.* [40] proposed an unmixing-based shadow correction method, which used unshadowed endmember spectra and abundance fractions to remove shadows. Zhang *et al.* [41] used a nonlinear spectral unmixing to detect shadow regions and restore information in shadow regions. These spectral unmixing-based shadow removal methods seriously rely on reasonable and accurate physical assumptions. Hartzell *et al.* [42] utilized light detection and ranging (LiDAR) spatial information for shadow detection of hyperspectral images, and designed a shadow restoration technique that leverages ratios derived from the overlapping LiDAR laser and hyperspectral image. Zhao *et al.* [43] proposed a cycle-consistent adversarial network for shadow compensation (SC-CycleGAN), in which the mapping function between shadow and nonshadow domains was learned by unpair images.

In this work, we model the shadow removal issue in hyperspectral images from a new perspective, i.e., multiexposure fusion. A multiexposure fusion-based shadow removal method is proposed for hyperspectral images, which mainly consists of three key steps. First, a color space conversion method is used to obtain the accurate shadow mask. Then, a set of multiexposure hyperspectral images is generated according to the principle of the intrinsic decomposition model. Finally, the generated multiexposure images are fused together with a two-stage image fusion scheme. Experiments on three real hyperspectral datasets indicate that the proposed method can obtain superior performance compared with other state-of-the-art shadow removal approaches. In general, the main contributions of this work are concluded as follows.

1) A multiexposure fusion-based shadow removal framework is proposed, which can effectively restore the spectral–spatial information in the shadow-corrupted regions. To the best of our knowledge, this article is the first work to investigate the shadow removal of hyperspectral images from the view of multiexposure fusion.
2) The principle of the intrinsic decomposition model is exploited to smartly generate a set of hyperspectral images with different exposure levels. The objective of this model is to estimate the spectral variations between shadow and nonshadow regions.
3) To better verify the effectiveness of the proposed method, we construct two hyperspectral shadow datasets, i.e., a mountain forest scene and a laboratory simulated dataset, which is beneficial for promoting the development of shadow removal methods in hyperspectral images.
4) Experiments on three hyperspectral datasets demonstrate that the proposed method can well restore the spectral–spatial information with respect to other methods. More importantly, the classification performance can be greatly improved after removing shadows.

This article is organized as follows. Section II briefly describes the intrinsic image decomposition. Section III shows the proposed method. Section IV is devoted to the experiment results. Finally, conclusions are reported in Section V.

II. Backgrounds

In 1978, the concept of "intrinsic decomposition" was first proposed according to Alhazen's observations [44]. In recent years, this problem has attracted great attention in image processing field. The "intrinsic decomposition" problem is modeled as the multiplication of the spectral reflectance component and the shading component

$$\mathbf{I} = \mathbf{R} \cdot \mathbf{S} \tag{1}$$

where $\mathbf{I}$ is the input hyperspectral image, $\mathbf{R}$ denotes the spectral reflectance component that reflects the material of different land covers, and $\mathbf{S}$ represents the shading component that is determined by the illumination of the scene.

III. Proposed Method

Fig. 1 shows the schematic of the proposed shadow removal method, which consists of three key steps: shadow detection, multiexposure image generation, and two-stage image fusion. The detailed steps are as follows.

A. Shadow Detection

To reduce the computing time of shadow detection, RGB bands selected from the original hyperspectral image (HSI) are used to detect the shadow regions. Specifically, a color space transformation method [45], [46] is used to detect shadow regions. First, the RGB bands (R: 625–740 μm, G: 492–577 μm, B: 440–475 μm) are extracted from the original images and converted to hue-saturation-intensity color space. Then, a shadow feature map is calculated according to hue and intensity components since shadow pixels have lower intensity and higher hue compared to the nonshadow regions

$$\mathbf{F} = \frac{\mathbf{H}^H + 1}{\mathbf{I}^H + 1} \tag{2}$$

where $\mathbf{H}^H$ and $\mathbf{I}^H$ represent the hue and intensity in hue-saturation-intensity color space, respectively. $\mathbf{F}$ is the shadow feature map, in which the pixel values in the shadow areas are higher than those in the nonshadow areas. According to this characteristic, an initial shadow detection map $\bar{\mathbf{A}}$ can be obtained by an adaptive threshold method

$$\bar{\mathbf{A}}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{F}_{i,j} > T \\ 0, & \text{Otherwise} \end{cases} \tag{3}$$

where T is the threshold value obtained by Otsu's method [47]. Finally, the final shadow mask $\mathbf{A}$ is obtained by using the attribute filter on the initial detection map $\bar{\mathbf{A}}$ to remove noisy labels.

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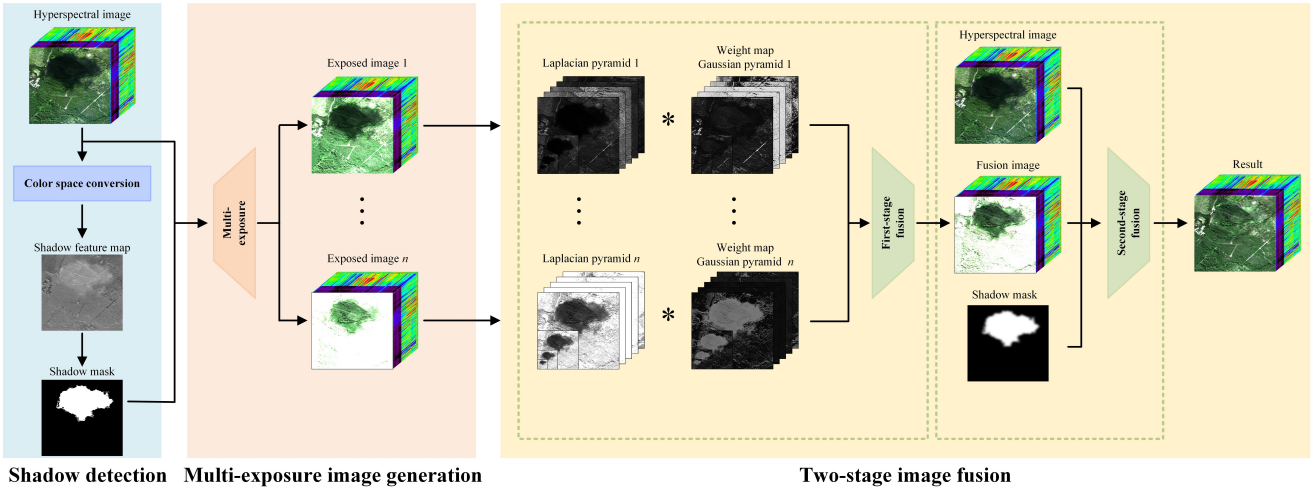

Fig. 1. Flowchart of the proposed shadow removal method.

B. Multiexposure Image Generation

Fusion of multiple exposure images provides much higher level of flexibility and can recover the shadow areas to have the same spectral reflectance of pixels with its shadow-free regions. The spectral reflectance-invariant characteristic of the same object between the shadow and nonshadow regions requires that the fusion should adaptively generate the multi-exposure images. In this work, we propose an intrinsic image decomposition-based multiexposure image generation method. Specifically, the hyperspectral image in the shadow regions is first extracted. Then, the exposure degree is produced according to the principle of the intrinsic decomposition model. Finally, the multiexposure images are obtained by pixel-wise multiplication of the shadow hyperspectral image and the exposure degree.

1) Hyperspectral Image Extraction in Shadow Regions: When the shadow mask is obtained, the hyperspectral image in shadow regions can be easily extracted

$$\mathbf{I}^s = \mathbf{A} \cdot \mathbf{I} \tag{4}$$

where $\mathbf{I}$ is the original hyperspectral image. $\mathbf{I}^s$ is the hyperspectral image in shadow regions, in which the spectral reflectance of shadow regions is preserved while the pixels in the nonshadow pixels are zero. "$\cdot$ " represents dot multiplication. $\mathbf{I}^s$ is dot multiplication result of the shadow mask $\mathbf{A}$ with each band of the original hyperspectral image $\mathbf{I}$.

2) Exposure Degree: To achieve the aim of shadow removal, we should estimate the exposure degree to make the shadow areas in the obtained multiexposure images have the same spectral reflectance as the nonshadow areas. First, the spectral reflectance difference of different bands between the shadow and nonshadow regions is calculated. To accurately estimate the spectral reflectance difference, the hyperspectral images inside and outside the shadow areas are extracted. An erosion operator is performed on the shadow mask

$$\mathbf{I}^r = (\mathbf{A} - (\mathbf{A} \ominus \mathbf{B})) \cdot \mathbf{I}^s \tag{5}$$

where $\mathbf{B}$ is a structural element, $\ominus$ represents the morphological corrosion operator, $\mathbf{I}^r$ represents the hyperspectral image in the shadow regions surrounding the shadow edges. The averaged spectral reflectance in the shadow regions can be estimated as

$$\mathbf{q}^r = \frac{\sum_{i=1}^{M} \mathbf{I}_i^r}{\sum_{i=1}^{M} (\mathbf{A}_i - (\mathbf{A}_i \ominus \mathbf{B}))} \tag{6}$$

where M denotes the total number of pixels and $\mathbf{q}^r$ is the averaged spectral reflectance in the shadow regions.

Accordingly, the averaged spectral reflectance in the nonshadow regions surrounding the shadow edges can be also estimated

$$\mathbf{I}_i^t = ((\mathbf{A} \oplus \mathbf{B}) - \mathbf{A}) \cdot \mathbf{I} \tag{7}$$

$$\mathbf{q}^t = \frac{\sum_{i=1}^{M} \mathbf{I}_i^t}{\sum_{i=1}^{M} ((\mathbf{A}_i \oplus \mathbf{B}) - \mathbf{A}_i)} \tag{8}$$

where $\mathbf{q}^t$ is the averaged spectral reflectance in the nonshadow regions surrounding the shadow edges, $\oplus$ represents the morphological dilation operator.

According to the principle of the intrinsic image decomposition, the spectral reflectance between the shadow and nonshadow regions is caused by the illumination component. Thus, the spectral reflectance difference of different bands between nonshadow and shadow regions can be expressed as

$$\mathbf{d} = \frac{\mathbf{R}^r \cdot \mathbf{S}^r}{\mathbf{R}^t \cdot \mathbf{S}^t} = \frac{\mathbf{q}^r}{\mathbf{q}^t} \tag{9}$$

where $\mathbf{R}^r$ and $\mathbf{R}^t$ are the spectral reflectance components in the nonshadow and shadow regions, respectively, $\mathbf{S}^r$ and $\mathbf{S}^t$ represent the illumination component in the nonshadow and shadow regions, respectively, $\mathbf{d}$ represents the spectral reflectance difference of different bands, which is obtained by dot division between $\mathbf{q}^r$ and $\mathbf{q}^t$.

Then, the exposure degree $\boldsymbol{\alpha}$ can be generated based on $\mathbf{d}$, which is expressed as

$$\boldsymbol{\alpha}_k = \boldsymbol{\lambda}_k \mathbf{d} \tag{10}$$

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where $\boldsymbol{\lambda}$ is a set of zoom factors, which is used to determine the number and intensity of exposed images.

3) Multiexposure Sequence Generation: The original HSI is exposed by multiplying the exposure degree $\boldsymbol{\alpha}$ to generate multiexposure hyperspectral images. In more detail, the multiexposure hyperspectral image sequence $\hat{\mathbf{I}}_k^o$ in the shadow regions is obtained as follows:

$$\hat{\mathbf{I}}_k^o = \boldsymbol{\alpha}_k \cdot \mathbf{I} \tag{11}$$

where the multiexposure image sequence $\hat{\mathbf{I}}_k^o$ is obtained by dot multiplication of exposure degree $\boldsymbol{\alpha}_k$ and hyperspectral images $\mathbf{I}$.

C. Two-Stage Image Fusion

In this section, a two-stage image fusion scheme is utilized to fuse the obtained multiexposure hyperspectral images $\hat{\mathbf{I}}_k^o$ so as to remove the shadows. In the first stage, a pyramid image fusion method is performed on the multiexposure hyperspectral images in the shadow regions to obtain shadow-free image. Specifically, the weight map is first calculated by using a Gauss curve

$$\mathbf{W}_{ij,k}^o = \exp\left(-\frac{\hat{\mathbf{I}}_{ij,k}^o - 0.5}{2\sigma^2}\right) \tag{12}$$

where the subscript ij,k is the pixel (i, j) in the kth hyperspectral image sequence, σ is a fixed coefficient, which is set as 0.2. To ensure the consistency of the fusion result, the K weight maps are normalized to one at each pixel (i, j)

$$\hat{\mathbf{W}}_{ij,k}^o = \left[\sum_{k=1}^{K} \mathbf{W}_{ij,k}^o\right]^{-1} \mathbf{W}_{ij,k}^o. \tag{13}$$

To avoid the seam problem that may be caused by using the discontinuous weight maps, a pyramid image fusion method is used to address this problem [48]. Given the lth level in a Laplacian pyramid decomposition of a hyperspectral image $\hat{\mathbf{I}}_k^o$ be regarded as $L\{\hat{\mathbf{I}}_{ij,k}^o\}^l$, and $G\{\hat{\mathbf{W}}_{ij,k}^c\}^l$ represents the Gaussian pyramid of the weight map $\hat{\mathbf{W}}_{ij,k}^o$. Then, the pyramid fused image in shadow regions can be expressed as

$$L\{\mathbf{P}_{ij,k}\}^l = \sum_{k=1}^{K} L\{\hat{\mathbf{I}}_{ij,k}^o\}^l G\{\hat{\mathbf{W}}_{ij,k}^o\}^l \tag{14}$$

where $L\{\mathbf{P}_{ij,k}\}^l$ is a Laplacian pyramid of the fused image. Finally, the multiexposure fused image $\mathbf{P}_{ij,k}$ can be obtained by inverse Laplacian pyramid transform on $L\{\mathbf{P}_{ij,k}\}^l$.

In the second stage, in order to make the shadow-free hyperspectral image more natural in the edges, the shadow mask $\mathbf{A}$ is first blurred by Gaussian filtering to obtain a soft shadow mask $G\{\mathbf{A}_{i,j}\}$. Then, the desired hyperspectral image is constructed by

$$\mathbf{Q}_{i,j} = \mathbf{P}_{i,j} G\{\mathbf{A}_{i,j}\} + \mathbf{I}_{i,j}\left(1 - G\{\mathbf{A}_{i,j}\}\right) \tag{15}$$

where $\mathbf{Q}$ is the final shadow removal result.

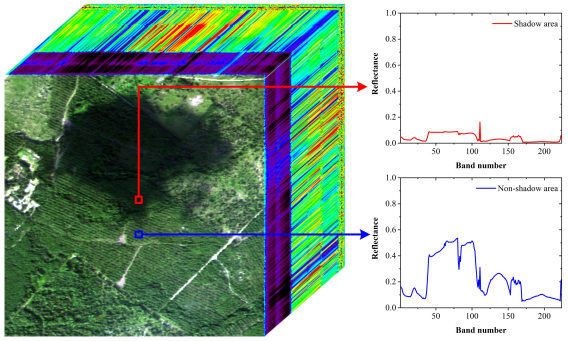

Fig. 2. Mountain forest dataset.

IV. Experiments

To examine the shadow removal effect of the proposed method, three hyperspectral datasets are adopted, including a mountain forest scene, a laboratory simulation scene, and an urban scene. First, the used datasets are described. Then, the experimental setup is given. Finally, the experimental results are discussed.

A. Datasets

In the experiments, three hyperspectral datasets from different scenes are utilized to evaluate the performance of the proposed method, which are described as follows.

1) *Mountain Forest Dataset (Mountain Forest Scene):* The mountain forest hyperspectral dataset was taken by the advanced equipment airborne visible infrared imaging spectrometer (AVIRIS) sensor, which contains 224 bands ranging from 0.4 μm to 2.5 μm and a spatial resolution of 20 m/pixel. The spatial size is 425 $\times$ 370. Fig. 2 shows the hyperspectral cube of the Mountain forest dataset.
2) *Simulated Dataset (Laboratory Simulated Scene):* The simulated dataset was collected in the laboratory by the GaiaField Hyperspectral sorter which has a push-broom imager with a spectral coverage range of 0.4 μm to 1 μm and a spectral resolution of 3.5 nm. The collected simulated dataset contains 176 bands of size 1546 $\times$ 960 pixels. As shown in Fig. 3(a), it is the corrupted hyperspectral image with shadow, in which shadow regions are formed by blocking some areas of light with baffles. Fig. 3(b) is the hyperspectral image without shadow collected by removing baffles. To eliminate the influence of the camera's dark current noise during the acquisition process, black-white normalization is used to process the collected raw data. The specific correction algorithm is that the raw data is normalized with the black frame and white frame data, which can be expressed as

$$\mathbf{X}_{\text{ref}} = \frac{\mathbf{X}_{\text{raw}} - \mathbf{X}_{\text{dark}}}{\mathbf{X}_{\text{white}} - \mathbf{X}_{\text{dark}}} \tag{16}$$

where $\mathbf{X}_{\text{raw}}$ is raw data, $\mathbf{X}_{\text{white}}$ is a single frame of the white image, which is collected by the imager aimed at a whiteboard, $\mathbf{X}_{\text{dark}}$ is a single frame of the black image,

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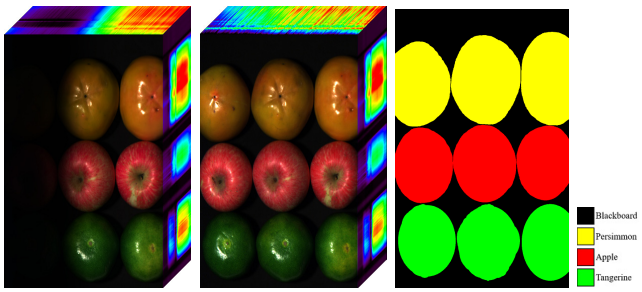

Fig. 3. Simulated dataset. (a) Simulated hyperspectral image corrupted by shadow. (b) Clean hyperspectral image. (c) Reference image. (d) Class name.

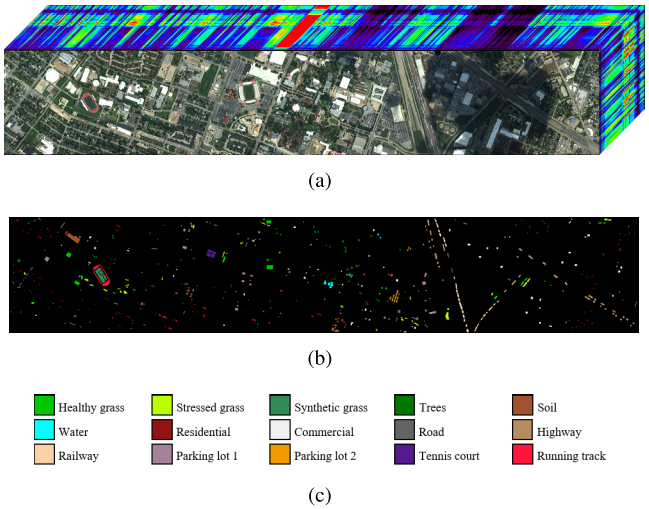

Fig. 4. Houston 2013 dataset. (a) Hyperspectral image. (b) Reference image. (c) Class name.

which is collected by covering the lens with its cap, $\mathbf{X}_{\text{ref}}$ is the corrected dataset after black-white normalization.

3) *Houston Dataset (Urban Scene):* The Houston hyperspectral dataset was distributed by the 2013 Data Fusion Contest of the IEEE Geoscience and Remote Sensing Society(GRSS), which was collected by the compact airborne spectrographic imager over the university of Houston campus and the neighboring urban area on June 23, 2012. This image consists of 144 bands and its spatial size is 349 × 1905 pixels with a spatial resolution of 2.5 m. The spectral coverage ranges from 0.38 μm to 1.05 μm. As shown in Fig. 4, the reference image contains 15 classes of interest.

B. Experimental Setup

1) Classification Evaluation Metrics: To quantitatively assess the performance of the proposed shadow removal method, three widely used indicators are adopted to measure the classification accuracy of shadow removal results, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient. OA represents the percentage of correctly classified pixels out of all pixels. AA indicates the average of the percentage of the correctly identified pixels for each class. Kappa coefficient is an index for consistency test, which is the

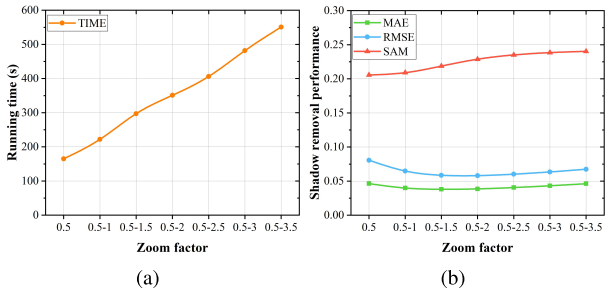

Fig. 5. Influence of the parameter $\boldsymbol{\lambda}$ on the running time and shadow removal performance of the proposed method. (a) Parameter $\boldsymbol{\lambda}$ to the influence of running time. (b) Parameter $\boldsymbol{\lambda}$ to the influence of shadow removal performance (MAE, RMSE, SAM).

percentage of correctly classified pixels corrected by a random agreement number. Ten experiments with randomly selected training samples are performed to obtain the mean values of the reported OA, AA, and Kappa coefficients.

2) Spectral Similarity Metrics: To evaluate the effect of the shadow removal, three widely used spectral similarity indexes are adopted, including mean absolute error (MAE), root mean square error (RMSE), and spectral angle mapper (SAM). MAE denotes the mean of the absolute error between the restored image and the reference image. RMSE is the standard deviation between the restored value and the true value. SAM measures the similarity of spectral curves by calculating the angle between restored spectral vectors and true spectral vectors. Lower values of MAE, RMSE, and SAM indicate better shadow removal performance.

3) Compared Approaches: In this work, three state-of-the-art shadow removal approaches are used as competitors, including multiscale retinex (MSR) [49], shadow compensation via cycle-consistent adversarial network (SC-CycleGAN) [43], dual generative adversarial networks (DualGAN) [50], a shadow removal method using color-lines (CL) [51], and interactive shadow removal (ISR) [52]. For the MSR method, parameters are set to default values in the original article. Each channel image of hyperspectral images is enhanced brightness by the Retinex algorithm. For the SC-CycleGAN method, 1×10^5 pixels are selected as training samples from shadow and nonshadow areas, respectively. Other parameters in this model, the learning rate, the batch size, etc. are set to be the default parameters in the corresponding publication. For the DualGAN method, the day-night translation pretrained model provided in the original publication is used to adjust the brightness of each channel of hyperspectral dataset. For the CL method, a linear shadow image model is used to estimate the color and illumination of hyperspectral images for each channel to produce shadow-free images. For the ISR method, an on-the-fly learning approach is used to detect shadows in images. Shadow removal is achieved by using a normalized frame to estimate shadow illumination changes of hyperspectral images for each channel.

4) Parameter Analysis: In this section, the zoom factor $\boldsymbol{\lambda}$ to the shadow removal performance of the proposed method is analyzed. An experiment is performed on the simulated dataset, in which $\boldsymbol{\lambda}$ is set to sets of vectors between

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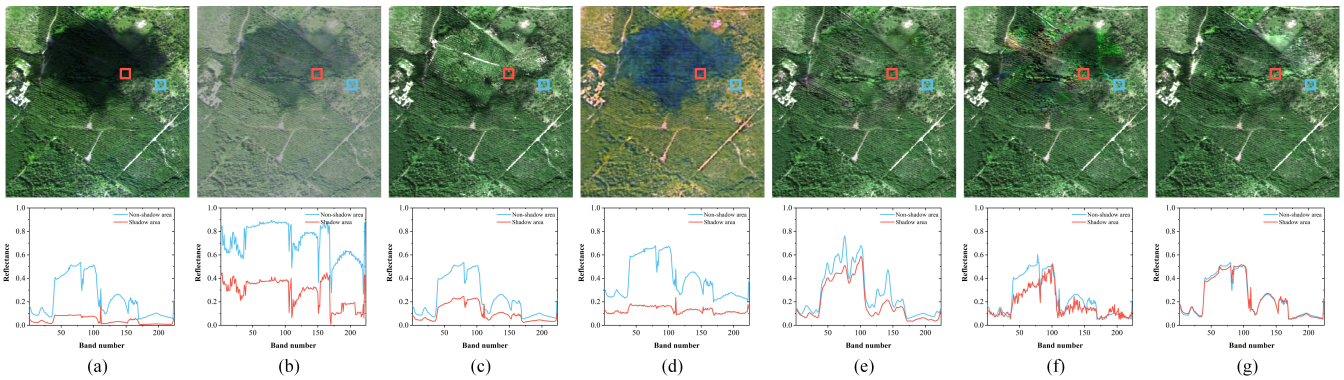

Fig. 6. Mountain forest image before and after removing shadow. (a) Original image. (b) MSR [49]. (c) SC-CycleGAN [43]. (d) DualGAN [50]. (e) CL [51]. (f) ISR [52]. (g) Our method. The blue curve is selected from the pixel (181, 310), and the red curve is selected from the pixel (156, 239).

TABLE I
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SIMULATED DATASET

Class Name	Training	Test	Classification accuracies of different methods (in %)						
			Original image	MSR	SC-CycleGAN	DualGAN	CL	ISR	Our method
Blackboard	100	476239	94.93	94.73	**96.73**	84.88	82.83	80.40	95.29
Persimmon	100	384495	65.75	84.00	62.91	84.40	84.61	93.51	**95.93**
Apple	100	321815	72.91	77.19	70.48	82.54	70.25	95.41	**96.92**
Orange	100	301211	87.73	78.45	63.35	86.94	90.61	92.27	**98.72**
OA			81.13	84.84	75.50	84.66	82.14	89.46	**96.51**
AA			80.33	83.60	73.37	84.69	82.08	90.40	**96.72**
Kappa			74.12	79.53	66.07	79.23	75.92	85.88	**95.30**

0.5 and 3.5 with an interval of 0.5. Fig. 5(a) shows the effect of the range of λ on the running time. It can be observed that as the range of λ increases, the running time gradually increases. Besides, Fig. 5(b) shows the influence of the parameter λ to the shadow removal performance. It is obvious that our method obtains the lower MAE, RMSE, and SAM when λ is set as [0.5, 1, 1.5] or [0.5, 1, 1.5, 2]. Therefore, λ is set to be [0.5, 1, 1.5, 2] as the default parameter for all datasets.

C. Result Analysis

1) Mountain Forest Scene: The first experiment is tested on the Mountain forest dataset. Fig. 6 shows resulting images obtained by different shadow removal methods. It is easily found that the MSR method can enhance the brightness of shadow areas. However, the nonshadow areas in the result are overexposed. For the SC-CycleGAN method, the shadows in the boundaries between the shadow and nonshadow regions cannot be effectively removed. The DualGAN method cannot recover the spatial details in the shadow regions and distort the truth colors of different land covers. For the CL method, the objects in the shadow areas can be observed, but the shadows cannot completely removed. The ISR method does not restore spectral information well and causes color changes in the shadow areas. By contrast, the proposed method can effectively remove the shadows and recover the finer details. In addition, the spectral curves of the same ground object in shadow and nonshadow areas are very similar, which also illustrates that the proposed method can well restore the spectral–spatial information.

TABLE II
SHADOW REMOVAL EFFECT OF DIFFERENT METHODS ON THE SIMULATED DATASET

Indexes	Best	MSR	SC-CycleGAN	DualGAN	CL	ISR	Our method
MAE	0	0.5326	0.0591	0.1730	0.0980	0.0917	**0.0384**
RMSE	0	0.5567	0.1044	0.2364	0.1543	0.1424	**0.0578**
SAM	0	0.3661	0.2742	0.2556	0.4429	0.6518	**0.2073**

2) Laboratory Simulated Scene: The second experiment is performed on the simulated dataset. The false-color images obtained by all shadow removal methods are shown in Fig. 7. It can be observed that the MSR method greatly enhances the image brightness. However, there is serious color distortion in the resulting image. The SC-CycleGAN method, the DualGAN method, and the ISR method fail to restore color information of different objects in the shadow areas. For the CL method, the color of different land covers in both shadow and nonshadow areas is distorted. By contrast, our method can well restore the true color of objects in shadow regions. Moreover, the reconstructed image of our method is closest to the clean hyperspectral image.

To illustrate the effectiveness of the shadow removal methods for object classification, the support vector machine (SVM) method [53] is used to evaluate the classification accuracy of different resulting images, in which 100 samples per class are randomly chosen as training set. Fig. 8 depicts the classification maps of the SVM method on different shadow removal images. By comparing Fig. 8(a) and (b), it can be seen that the objects in the shadow regions cannot be correctly

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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET

Class Name	Training	Test	Classification accuracies of different methods (in %)						
			Original image	MSR	SC-CycleGAN	DualGAN	CL	ISR	Our method
Healthy grass	187	1064	86.18	**98.97**	97.27	90.32	98.50	97.56	94.83
Stressed grass	188	1066	98.97	98.78	98.41	92.03	97.65	**99.16**	98.59
Synthetic grass	104	593	99.49	98.65	88.03	96.63	**100.00**	99.66	99.33
Trees	186	1058	98.11	95.65	93.19	89.41	98.39	98.39	**98.58**
Soil	186	1056	98.96	98.20	98.86	97.06	**99.15**	99.05	98.39
Water	48	277	83.39	97.83	92.06	70.76	77.62	**98.19**	96.75
Residential	190	1078	87.76	89.80	77.92	78.01	**94.90**	89.98	88.40
Commercial	186	1058	51.42	**70.89**	63.14	56.90	67.11	70.51	59.55
Road	187	1065	71.64	70.99	72.68	58.59	**78.78**	72.96	73.15
Highway	184	1043	81.59	66.35	81.50	60.59	66.83	80.35	**85.81**
Railway	185	1050	70.76	80.29	65.62	59.90	**90.48**	86.19	88.57
Parking Lot 1	184	1049	64.73	60.06	63.11	**79.98**	49.86	40.51	67.97
Parking Lot 2	70	399	8.02	**35.59**	27.07	22.31	14.79	23.06	24.06
Tennis Court	64	364	96.43	**98.35**	96.15	72.80	93.68	97.25	97.53
Running Track	99	561	99.29	98.75	98.93	91.62	98.93	**99.47**	98.57
OA			80.92	83.74	81.27	76.03	83.58	83.80	**85.31**
AA			79.78	83.94	80.93	74.46	81.78	83.49	**84.67**
Kappa			79.34	82.42	79.74	74.02	82.21	82.46	**84.10**

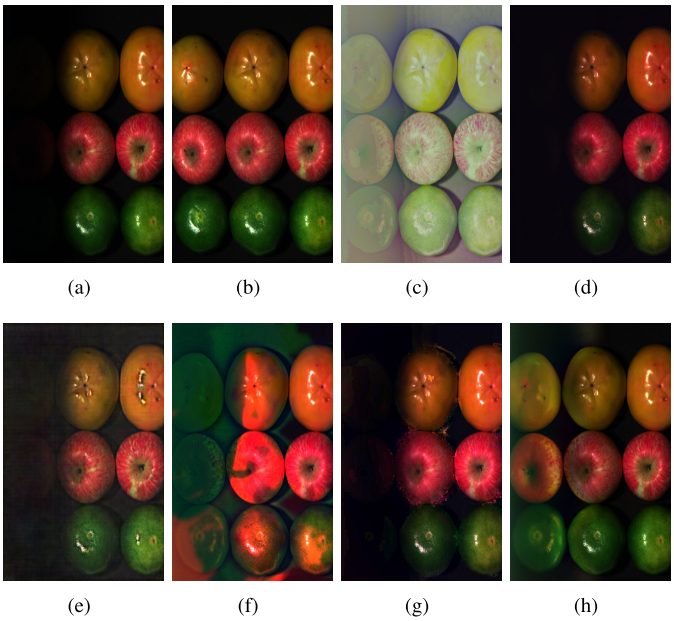

Fig. 7. Simulated image before and after removing shadow. (a) Original image. (b) Clean image. (c) MSR [49]. (d) SC-CycleGAN [43]. (e) DualGAN [50]. (f) CL [51]. (g) ISR [52]. (h) Our method.

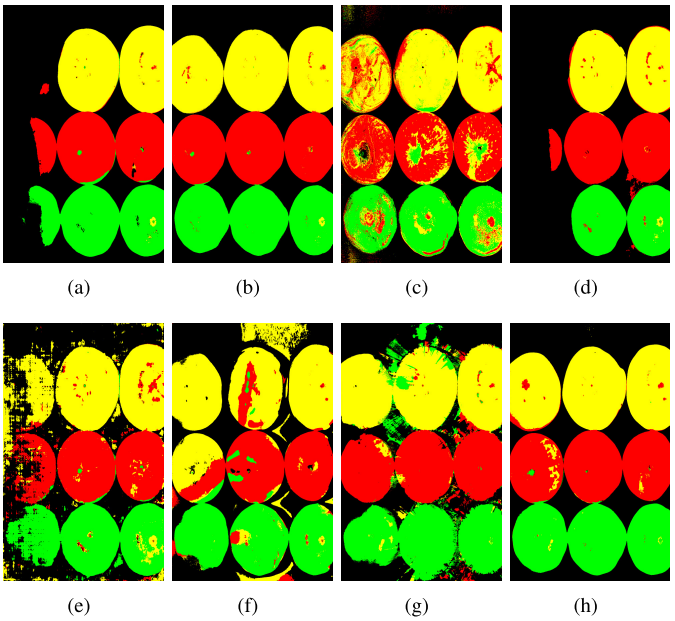

Fig. 8. Classification maps of the SVM method on the shadow removal results of different methods. (a) Original image corrupted by the shadows, OA = 81.13%. (b) Clean image, OA = 98.24%. (c) MSR [49], OA = 84.84%. (d) SC-CycleGAN [43], OA = 75.50%. (e) DualGAN [50], OA = 84.66%. (f) CL [51], OA = 84.47%. (g) ISR [52], OA = 89.46%. (h) Our method, OA = 96.51%.

classified. The reason is that the spectra of different objects in shadow regions have been changed. The MSR method yields a very noisy classification map. This is due to the fact that the spectra in shadows cannot be recovered by image enhancement technique. For the SC-CycleGAN method, the objects in the shadow regions cannot be identified since the SC-CycleGAN method fails to remove the interference of the shadows. For the DualGAN method, the classification map appears noisy spots, especially for the shadow regions. For the CL method, although the objects in shadow areas can be detected, most of them are misclassified. The ISR method results in a lot of noisy spots in the nonshadow areas. Different from these methods, our method can greatly boost the classification performance in the shadow regions. The classification map of our method is more similar to the classification map of the clean image. Furthermore, the objective indexes of all classification results are presented in Table I. It can be observed that our method provides the highest values in terms of OA, AA, and Kappa coefficient. This also confirms that our method significantly outperforms other shadow removal methods in improving the classification accuracy.

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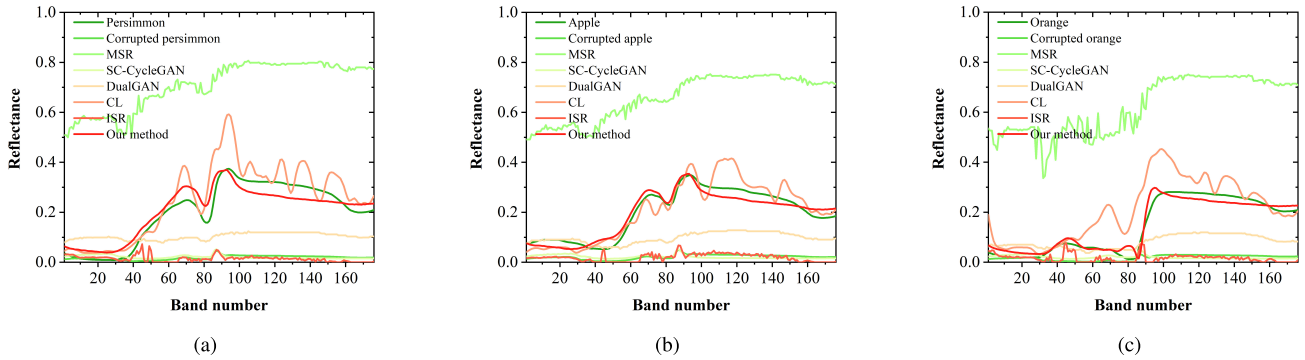

Fig. 9. Spectral curves of three objects at the same position. (a) Persimmon (520, 134). (b) Apple (951, 226). (c) Orange (1186, 245).

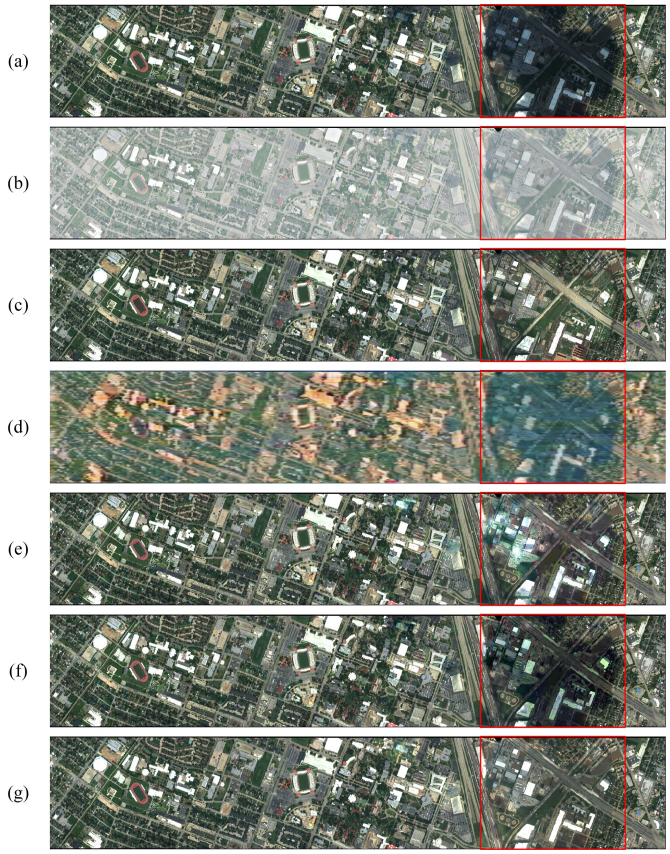

Fig. 10. Shadow removal results obtained by different methods on the Houston dataset. (a) Original image. (b) MSR [49]. (c) SC-CycleGAN [43]. (d) DualGAN [50]. (e) CL [51]. (f) ISR [52]. (g) Our method.

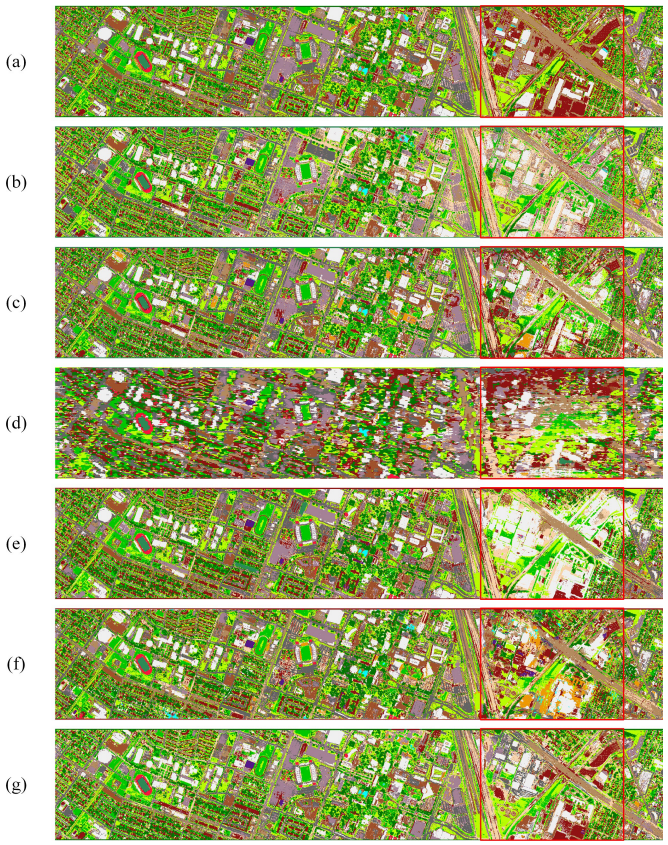

Fig. 11. Classification maps of different methods on the Houston dataset. (a) Original image, OA = 80.92%. (b) MSR [49], OA = 83.74%. (c) SC-CycleGAN [43], OA = 81.27%. (d) DualGAN [50], OA = 76.03%. (e) CL [51], OA = 83.59%. (f) ISR [52], OA = 83.80%. (g) Our method, OA = 85.31%.

In addition, Fig. 9 shows the spectral curves of different objects at the same position in shadow regions. As shown in Fig. 9, the SC-CycleGAN method, the DualGAN method, and the ISR method cannot well restore the spectra of three objects. The CL method changes the spectral reflectance of some objects, which is significantly different from the clean spectral reflectance. The MSR method greatly increases the spectral reflectance of shadow pixels. By contrary, our method has the best spectral recovery ability. The reconstructed spectral curves of three objects in the shadow regions are closest to their clean spectral curves. Furthermore, Table II lists the objective metrics of spectral reconstruction by different methods. Compared with other methods, the proposed method obtains the lowest MAE, RMSE, and SAM, which indicates that the shadow removal image obtained by our method is closest to the clean image.

3) Urban Scene: The third experiment is conducted on the Houston dataset. Fig. 10 gives false-color images of different shadow removal methods. It can be easily seen that the MSR method and the SC-CycleGAN method can enhance the brightness of shadow areas. However, the MSR method yields

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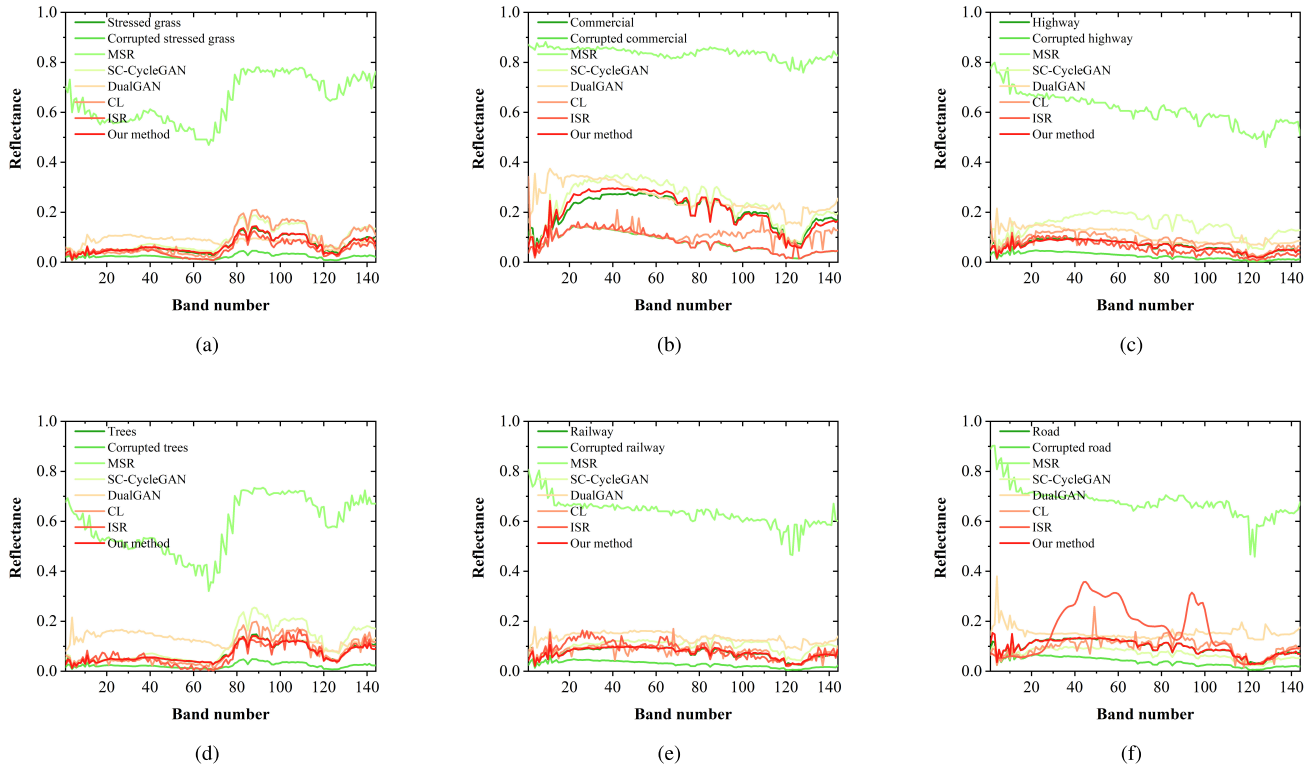

Fig. 12. Spectral curves of different methods for six objects. (a) Stressed grass (201, 1551). (b) Commercial (283, 1570). (d) Highway (207, 1691). (d) Trees (177, 1721). (e) Railway (246, 1479). (f) Road (28, 1395).

an overexposure phenomenon, and the SC-CycleGAN method cannot well remove the shadows in the boundaries between the shadow and nonshadow regions. The DualGAN method suffers from serious spectral distortion. The CL method and the ISR method cannot remove shadows from the original images. By contrast, our method can effectively recover the spatial details of different ground objects in the shadow regions. Moreover, there is no distinct demarcation line on the border of shadow regions, and the color of different objects in the shadow areas is more similar with one of the nonshadow areas.

Fig. 11 presents the classification maps of the SVM method on different shadow removal results. The training set which accounts for 15% is randomly selected from the reference image. It can be seen that the MSR method and the SC-CycleGAN method can slightly enhance the classification performance of ground objects in shadow areas. However, the MSR method cannot well classify the residential class in the shadow regions. The SC-CycleGAN method yields serious misclassification in the edge regions. The DualGAN method produces unsatisfactory classification performance, since the spectral-spatial information in the recovered dataset has been seriously distorted. The CL method severely results in the misclassification of some ground objects in shadow regions because their spectra are altered. The ISR method cannot well identify the ground objects in the shadow areas. By comparison, our method can increase the classification performance of different objects in the shadow regions, such as houses, roads, and trees. Besides, Table III lists the classification performance of all methods. It can be easily observed that our method yields the highest classification accuracy in terms of OA, AA, and Kappa coefficient.

In addition, the spectral curves of different objects are shown in Fig. 12, in which the positions of these clean objects, i.e., stressed grass, commercial, highway, trees, railway, and road, are located at (112, 1114), (184, 1078), (265, 1801), (16, 1863), (26, 1294), and (22,1298), respectively. The MSR method severely amplifies the spectral reflectance of different ground objects in the shadow areas. The reconstructed spectral reflectance of different objects in the shadow areas is always higher than those in the nonshadow areas. The DualGAN method cannot restore the pure spectral information of different land covers. The SC-CycleGAN method slightly improves the reconstructed effect of spectral curves than other methods. However, the spectral reflectance of some objects such as highways and trees is still unsatisfactory. The spectral curves recovered by the CL method and the ISR method are quite different from the clean spectral curves on some objects, such as commercial and roads. Among all considered methods, the spectral curves of different objects obtained by our method are very close to that of different objects in the nonshadow regions. This experiment also demonstrates that the proposed method can effectively remove the shadows in the source image.

D. Computing Cost

All experiments are implemented with MATLAB 2018b on a desktop computer configured with Intel (R) Core (TM) i9-10900K central processing unit (CPU) processor (3.70 GHz), 64 GB memory, and 64-bit operating system. The running time of different shadow removal methods on three hyperspectral datasets is shown in Table IV. It is found that our method is computationally efficient on three hyperspectral datasets

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TABLE IV
COMPUTING TIME OF DIFFERENT METHODS

Datasets	Running time of all studied methods (in seconds)					
	MSR	SC-CycleGAN	DualGAN	CL	ISR	Our method
Mountain forest	116.14	165005.04	10178.23	230.42	977.26	**86.84**
Simulated image	755.52	159025.29	10317.47	2411.24	11908.19	**354.77**
Houston	214.66	156009.37	10194.85	846.85	5235.09	**152.64**

among all studied methods, which is more suitable for practical applications. The reason is that the proposed method only requires several efficient operations, such as pyramid decomposition, dot multiplication, and dot division.

V. CONCLUSION

In this work, we propose a multiexposure fusion-based shadow removal framework for hyperspectral images, which mainly includes three key steps. First, a color space-transformed shadow detection method is utilized to obtain shadow masks. Then, the intrinsic decomposition model is explored to compute exposure degree for generating a sequence of multiexposure images. Finally, a two-stage image fusion method is utilized to remove image shadows. Experiments performed on three hyperspectral datasets verify that our method achieves the best shadow removal performance compared to other state-of-the-art hyperspectral shadow removal methods. Moreover, the classification accuracy can be greatly increased after removing shadows in hyperspectral images.

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